

- [The Yoneda Lemma: Central Theorem of Category Theory](#)
 - [Statement](#)
 - [Interpretation](#)
 - [Proof Sketch](#)
 - [Four Equivalent Formulations](#)
 - [1. Hom-Set Version \(Most Common\)](#)
 - [2. Representation Version](#)
 - [3. Natural Transformation Version](#)
 - [4. Embedding Version \(Yoneda Embedding Corollary\)](#)
 - [Why This Is True \(The Deep Reason\)](#)
 - [12 Critical Examples](#)
 - [1. Forgetful Functor is Representable](#)
 - [2. Free Functor is Represented](#)
 - [3. Power Set is Representable](#)
 - [4. Fundamental Group Functor](#)
 - [5. Homology Functors](#)
 - [6. Tangent Space in Manifolds](#)
 - [7. Scheme Points](#)
 - [8. Tensor Product is Representable](#)
 - [9. Determinant as Representable Functor](#)
 - [10. Moduli Spaces as Representing Objects](#)
 - [11. Stone Duality](#)
 - [12. Eilenberg-MacLane Spaces](#)
 - [Three Consequences](#)
 - [Consequence 1: Density Theorem](#)
 - [Consequence 2: Functors Determine Each Other via Hom](#)
 - [Consequence 3: The Yoneda Embedding is Dense](#)
 - [The Converse \(Co-Yoneda\)](#)
 - [Study Checklist](#)
 - [Key Insight](#)
 - [Relation to Kan Extensions](#)

The Yoneda Lemma: Central Theorem of Category Theory

Statement

Theorem (Yoneda Lemma): For any functor $F: \mathcal{C}^{\text{op}} \rightarrow \text{Set}$ and any object $A \in \mathcal{C}$:

$$\text{Nat}(\text{Hom}(-, A), F) \cong F(A)$$

The isomorphism is **natural in both F and A** .

Interpretation

In English: “Natural transformations from the Hom-functor to F are in bijection with elements of $F(A)$.”

More precisely: Evaluating a natural transformation η at the identity element id_A gives you the element $\eta_A(\text{id}_A) \in F(A)$, and this map is bijective.

Proof Sketch

1. **Forward direction:** $\eta_A: \text{Hom}(-, A) \rightarrow F(A)$
 - Given natural transformation $\eta: \text{Hom}(-, A) \rightarrow F$
 - Evaluate at A : $\eta_A: \text{Hom}(A, A) \rightarrow F(A)$
 - Apply to identity: $\eta_A(\text{id}_A) \in F(A)$
2. **Reverse direction:** $x \mapsto \eta_x$ natural transformation $\eta: \text{Hom}(-, A) \rightarrow F$
 - Given element $x \in F(A)$
 - For each $f: B \rightarrow A$ and $y \in \text{Hom}(B, A)$, define:

$$\eta_x(y) = F(f)(x)$$
 - This is natural: commutes with further morphisms
3. **Bijection verification:** The forward-backward compositions are identities.

Four Equivalent Formulations

1. Hom-Set Version (Most Common)

$\text{Nat}(\text{Hom}(A, -), F) \cong F(A)$
 $\eta \mapsto \eta_A(\text{id}_A)$

2. Representation Version

For any functor $F: \mathcal{C} \rightarrow \mathbf{Set}$:

F is representable $\iff \exists A \in \mathcal{C}$ such that $F \cong \text{Hom}(A, -)$

3. Natural Transformation Version

Every natural transformation $\eta: \text{Hom}(-, A) \rightarrow F$ is uniquely determined by $\eta_A(\text{id}_A)$.

4. Embedding Version (Yoneda Embedding Corollary)

The Yoneda embedding:

$y: \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$
 $X \mapsto \text{Hom}(-, X)$

is **fully faithful**: For any $A, B \in \mathcal{C}$:

$\text{Hom}_{\mathcal{C}}(A, B) \cong \text{Nat}(\text{Hom}(-, A), \text{Hom}(-, B))$

Why This Is True (The Deep Reason)

The Yoneda lemma is true because **morphisms uniquely determine functors**.

When you have a natural transformation $\eta: \text{Hom}(-, A) \rightarrow F$, the naturality square:

$$\begin{array}{ccc}
 \text{Hom}(B, A) & \xrightarrow{\quad} & \cdot_B \quad F(B) \\
 \text{post-comp} \quad F(\text{post-comp}) & & \\
 \downarrow & & \downarrow \\
 \text{Hom}(A, A) & \xrightarrow{\quad} & \cdot_A \quad F(A)
 \end{array}$$

forces $\cdot_B$ to be completely determined by what $\cdot$ does at $\text{id}_A \in \text{Hom}(A, A)$.

This is because **every morphism $f: B \rightarrow A$ can be expressed as post-composition of id_A with something**, and naturality forces $\cdot_B(f)$ to be determined by $\cdot_A(\text{id}_A)$.

12 Critical Examples

1. Forgetful Functor is Representable

$F: \text{Grp} \rightarrow \text{Set}$ (forgetful) $E \in \text{Hom}(\$, -)$ Elements of $F(G) = \text{homomorphisms from } \$ = \text{elements of } G$

2. Free Functor is Represented

$F: \text{Set} \rightarrow \text{Grp}$ (free group) has right adjoint (forgetful) So F is left adjoint to $\text{Hom}(F(-), -)$

3. Power Set is Representable

$P(X) \in [X, \{0,1\}]$ Subsets of X functions to two-element set

4. Fundamental Group Functor

$\pi_1: \text{Top}^* \rightarrow \text{Grp}$ Not representable (why? would need universal space with specific homotopy properties)

5. Homology Functors

$H_n: \text{Top} \rightarrow \text{Ab}$ Not representable (Eilenberg-MacLane spaces represent these)

6. Tangent Space in Manifolds

$T_p M \in (m_p / m_p^2)^*$ where $m_p = \text{ideal of functions vanishing at } p$ This is representability in derived setting

7. Scheme Points

$\text{Spec}(R): \text{Hom}(-, \text{Spec}(R))$ represents points of scheme Yoneda: points are functors from finitely-generated algebras

8. Tensor Product is Representable

$M \otimes_R N$ represents: $\text{Hom}(-, M \otimes_R N) \in \text{Bilinear maps}$

9. Determinant as Representable Functor

$\det: \text{GL}_n(R) \rightarrow R^*$ is represented by the universal matrix

10. Moduli Spaces as Representing Objects

Moduli of curves M_g : $\text{Hom}(-, M_g) =$ families of curves over $-$ Yoneda: moduli spaces represent universal families

11. Stone Duality

$\text{Spec}(\text{Boolean algebra}) \cong$ representation by Boolean algebra operations Boolean algebras are representable in topological spaces

12. Eilenberg-MacLane Spaces

$K(G, n)$: $\text{Hom}(-, K(G, n)) \cong H^n(-, G)$ Cohomology is representable by topological spaces

Three Consequences

Consequence 1: Density Theorem

Every presheaf is a colimit of representables.

$$F \in [C^{\text{op}}, \text{Set}] = \text{colim}_{\{ \text{Hom}(c, -) \in F \}} \text{Hom}(c, -)$$

where the colimit is over the **category of elements** (c, x) where $c \in C$ and $x \in F(c)$.

Consequence 2: Functors Determine Each Other via Hom

If two functors $F, G: C^{\text{op}} \rightarrow \text{Set}$ satisfy:

$$\text{Hom}(Y_C(A), F) \cong \text{Hom}(Y_C(A), G) \text{ for all } A \in C$$

(where $Y_C(A) = \text{Hom}(-, A)$), then $F \cong G$.

This is because every presheaf is a colimit of representables, and Hom commutes with limits.

Consequence 3: The Yoneda Embedding is Dense

The image of $y: C \rightarrow [C^{\text{op}}, \text{Set}]$ is **dense**: every presheaf is a colimit of representables.

The Converse (Co-Yoneda)

Lemma (Duality): For contravariant $F: C^{\text{op}} \rightarrow \text{Set}$, similarly:

$$\text{Nat}(\text{Hom}(A, -), F) \cong F(A)$$

with the covariant version of composition.

This is why **left/right Kan extensions are dual**.

Study Checklist

- ☐ Understand the Yoneda lemma statement word-by-word

- ☐ Verify it in 3-4 concrete examples (representable functors)
- ☐ Understand why the bijection is natural
- ☐ Know the Yoneda embedding and why it's fully faithful
- ☐ Understand density theorem (colimit of representables)
- ☐ Connect to Kan extensions (representables as Kan extensions)

Key Insight

The Yoneda lemma says: **Morphisms and natural transformations freely determine functors.**

This is why Kan extensions exist: they're asking "What's the universal way to extend a functor?" The answer is determined by the Yoneda lemma structure.

Every representable functor is a **right Kan extension** of the identity functor. Understanding Yoneda deeply means understanding why Kan extensions work.

Relation to Kan Extensions

The Yoneda lemma is the **simplest case** of Kan extension theory: - $\text{Hom}(-, A)$ is obtained by Kan extending along the Yoneda embedding - The natural isomorphism in Yoneda is the universal property of this Kan extension - The density theorem (colimit of representables) is the **density theorem for Kan extensions**

In fact: **Yoneda lemma = Kan extension in the category $[C^{\text{op}}, \text{Set}]$.**

This is why mastering Yoneda is essential preparation for Kan extensions.